Combinatorics and counting

Lecture and worked examples

Maths Explorer

Objectives

Choose a counting model; distinguish order and repetition; use binomial coefficients and Pascal's rule.

Add or multiply

For pairwise disjoint finite sets A_i, $|\bigcup A_i| = \sum |A_i|$. If the cases overlap, the double counting must be corrected: $|A \cup B| = |A| + |B| - |A \cap B|$. The number of elements of the Cartesian product $A \times B$ is $|A||B|$. For a sequence of choices, multiply the numbers of possibilities at each step, provided these numbers are valid for every earlier choice considered.

Add or multiply

A code made of two letters chosen from 26, followed by three digits, with repetition allowed, can take $26^2 \times 10^3$ values. A three-digit number does not start with 0: there are 9×10^2 of them.

Lists and subsets

In a set of n elements, there are n^k ordered lists of length k with repetition; $n(n-1)\cdots(n-k+1) = \dfrac{n!}{(n-k)!}$ lists of k distinct elements, if $0 \leq k \leq n$; and $n!$ permutations of all the elements. We set $0! = 1$.

Lists and subsets

A combination of k elements from n is a subset with k elements; order does not matter.

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}, \quad \binom{n}{0} = \binom{n}{n} = 1, \quad \binom{n}{1} = n, \quad \binom{n}{2} = \frac{n(n-1)}{2}.$$

Lists and subsets

Each subset of size k gives exactly $k!$ distinct lists, according to the order of its elements. The number of lists without repetition is therefore $k!\binom{n}{k}$.

Identities and proofs

$$\binom{n}{k} = \binom{n}{n-k}, \qquad \binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}, \qquad \sum_{k=0}^{n} \binom{n}{k} = 2^n.$$

Pascal's rule above is used for $0 \le k < n$; the boundary formulas can be read directly from the values 1.

Identities and proofs

Taking the complement sets up a one-to-one correspondence between subsets of size k and those of size $n - k$.

Identities and proofs

In a set of $n+1$ elements, fix an element a. A subset of size $k+1$ either contains a (choose k other elements from n) or does not contain it (choose $k+1$ elements from n). The two cases are disjoint and cover all possibilities.

Identities and proofs

For $0 \le k < n$, put over a common denominator:

$$\frac{n!}{k!(n-k)!} + \frac{n!}{(k+1)!(n-k-1)!} = \frac{n!((k+1)+(n-k))}{(k+1)!(n-k)!} = \frac{(n+1)!}{(k+1)!(n-k)!}.$$

The boundary cases are handled directly.

Identities and proofs

For each of the n elements, choosing ”present” or ”absent” gives 2^n subsets. Sorting them by size, we count $\sum_{k=0}^{n} \binom{n}{k}$: the two counts are equal.

Identities and proofs

The expansion $(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k$ is understood by choosing, in each of the n factors, either the term a or the term b. This formula is not used as a prerequisite for the assessments in this booklet.

Research activity

Order and repetition in a counting problem - 30 min.

Six distinct students volunteer to organise a debate.

Research activity (2)

1. Two students are chosen to form a pair, with no roles assigned. How many pairs can be formed?
2. A chair and a secretary are appointed, and they must be two different students. How many assignments of the two roles are possible?
3. Three successive speeches are planned. The same student may speak several times. How many programmes of three speeches are possible?

Research activity (3)

4. Start again with only four students, labelled A, B, C and D. Write out all the pairs and all the assignments of the two roles. Then explain why there are 64 programmes of three speeches, without listing them.
5. For each of the three ways of choosing, state whether order matters and whether the same person can be chosen several times.